

IFRS 17: THE ACTUARY AS A STORYTELLER

ASTIN WEBINAR

31 JAN 2023

VASILKA BANGEOVA, GUY CARPENTER



1. New disclosure challenges for actuaries

2. Observations from recent non-life public disclosures

- Population for our study
- Key messages
- Policy choices
- Impact on KPIs

Agenda

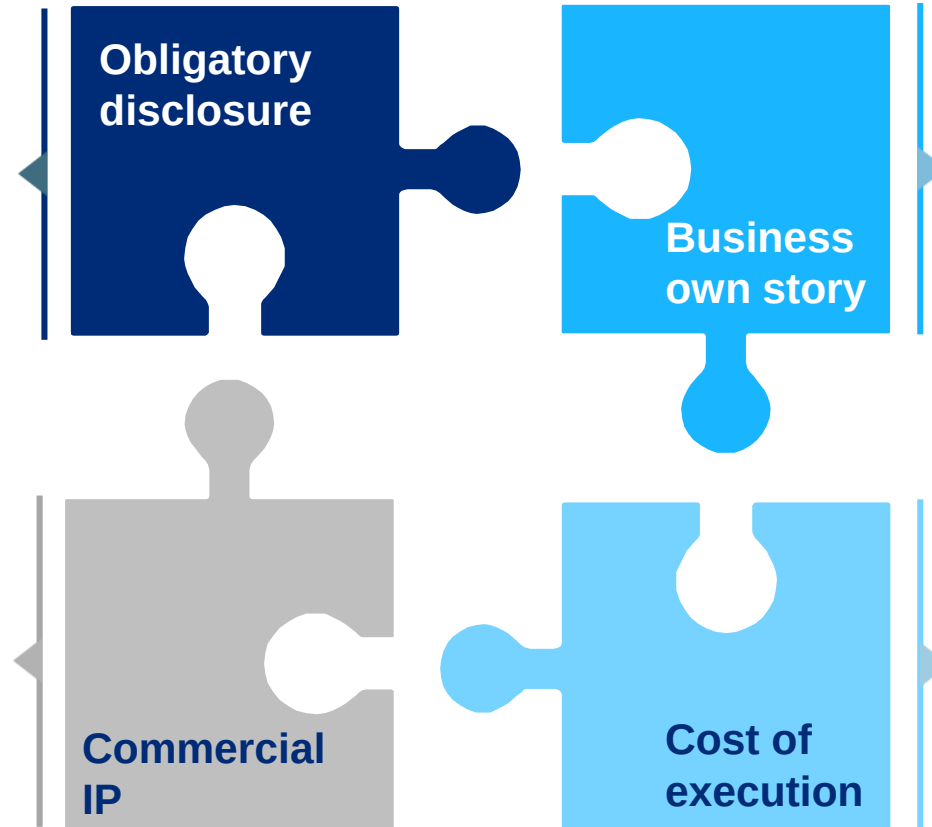
New disclosure challenges for actuaries



The actuary as a storyteller

New challenges under IFRS 17

- Numerous new and expanded disclosures (see next slide)
- Prescribed minimum content
- Stronger emphasis on judgements, estimates, inputs and process
- Holistic perspective (ALM view)
- A challenging balancing act:
 - Compliance
 - Credibility
 - Protection of commercially sensitive know-how



- Explain analysis of change
 - Growth / creation of value
 - Development of risk (AvE)
 - Assumptions or methodology changes
- Measure impact on CoR, RoE, Leverage, etc
- Recalculate KPIs on IFRS 17 basis and compare to IFRS 4
- Test impact of policy choices
- Use existing MI to translate 'old' into 'new'
- Test stakeholder reaction
- Redesign

A snapshot of what is new in disclosures

Selected points of focus for non-life actuaries

New	
PAA	How the company has satisfied eligibility requirements
Discount rates	Yield curve (or range of yield curves) used to discount cash flows
Risk adjustment	Confidence level and technique (if other) - justification and corresponding conf. level
IFIE	Total amount & explanation of relationship of IFIE and investment return on assets
CSM	(Quantitative) when the company expects to recognise CSM in P&L (in time bands)
Regulatory	Effects of the regulatory framework in which the company operates

Expanded	
Liquidity (Maturity analysis)	• Net cash flows (PV or undiscounted) for each of the first 5 years post reporting date, cumulative thereafter • Amounts payable on demand (and relationship to carrying amounts of contract portfolios)
Sensitivities	For each type of market risk (e.g. FX , interest rates) – explain relationship between sensitivities from insurance contracts and those from financial assets

Illustrative example extracts are provided in the Appendix.



...emphasis on the inputs, assumptions and estimation techniques used, and the processes for estimating the inputs to measurement methods

Population for our study

2

Population of preliminary public disclosures

Measurement

Global and International Composites



- Significant Life business impacts Group metrics via CSM
- P&C measured mostly under the simplified PAA approach (some exceptions: Talanx, SCOR)
- Assumed reinsurance and retro(cessions) measured under a mixture of GMM and PAA (small CSM)
- Transition at retrospective or fair value approach (for assumed reinsurance and some run-off portfolios)

International and Local Non-life



- Business measured under the simplified PAA approach
- Transition (where disclosed) at full retrospective approach

...all disclosures per unaudited information in investor presentations held prior to 31 Dec 2022

Key messages

3

IFRS 17 Transition: key messages in preliminary public disclosures



Strategy

- Business strategy is unaffected
- No major change expected in underwriting & investment risk appetite



Solvency

- Better alignment with Solvency II and internal steering basis
- No impact on solvency ratio (SII or SST)



Cash & Dividends

- No change to cash generation / remittances, dividends guidance or share buy-backs (where announced)



Performance metrics (KPIs)

- Key financial targets reaffirmed
- Traditional P&C KPIs remain in use but are calculated on IFRS 17 basis (devil in the detail)
- New adjustments (varying practices) for ROE and Leverage calculations



Equity & P&L

- Minimal equity effect on transition
- Equity: more stable (less sensitive to interest rate changes)
- P&L: changed mix of underwriting and finance results; more P&C volatility (depending on accounting choices)

Policy choices

4

IFRS 17 observed policy choices

Discount rates

Prevalent choices in the population for PAA

Bottom-up approach with illiquidity premium
Risk free rate aligned with SII



Change in discount rates goes to OCI

Nuanced application:



Illiquidity premium (IP) based on EIOPA replicating portfolios approach



Alignment with SII for P&C and GMM business; VFA IP based on own asset mix to reflect ALM



Risk free determined using swap curves for each settlement currency. IP by reference to corporate bond spreads



No illiquidity premium for PAA, risk free rate only



No illiquidity premium on onerous contracts

Different approach or policy choice:



Top-down method (to capture non-fundamental spread in own asset portfolio); changes to OCI



Not explicitly disclosed (changes appear to go to P&L)



Not explicitly disclosed (changes appear to go to P&L)



Change in the discount rate is recognised in IFIE in P&L

IFRS 17 and IFRS 9 observed policy choices

Aligning accounting outcomes with ALM objectives

Prevalent choices in the population (for PAA business)	
Fixed income assets	Measured at FVOCI (mark-to-market changes go to OCI) with recycling of realised gains / losses in P&L
Equities	Measured at FVOCI with no recycling of realised gains / losses in P&L
Funds investments	Measured at FVTPL (mark-to-market changes go to P&L)

Different policy choice - measured at FVTPL:

	for assets matching insurance liabilities			in recognition of underlying economics		for debt securities	
---	---	--	---	--	--	---------------------	--

No explicit disclosure:

		
---	---	---

IFRS 17 observed policy choices

Risk adjustment

Methodology and calibration

Confidence level



Set at 65th percentile with a range 62.5th - 67.5th percentile



ageas 75th percentile

tal anx.

65th – 75th percentile



Based on SST target level of $\geq 160\%$



Corridor around 90th percentile



c75th percentile



Calculation based on VaR, percentile not specified

Cost of capital (CoC)



6% CoC and more diversification than for SII



CoC < CoC for SII (undisclosed) and more diversification than for SII



4% CoC and more diversification than for SII

Partial or no disclosure



'Assumptions consistent with SII' (not specified)



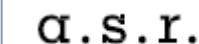
Reference to SII but calibrated to own view of risk and diversification (around 70th percentile)



No disclosure



No disclosure



No disclosure

Impact on KPIs

4

IFRS 17 impact on performance metrics

Return on equity

Direction of change	Calculation approach
Majority within the population expect: - Improved RoE under IFRS 17 (targets: low to mid-double digit) - Less volatility over time	'Standard' = using net shareholder income and equity values as reported in financial accounts, and/or 'Non-standard' = Using non-GAAP income measures and adjusted equity values

No explicit disclosure:



Examples:

	RoE = Underlying Earnings / Average Shareholders Equity (excl. OCI)
	RoE = Consolidated result / Adjusted equity (for unrealised G/L, cash flow hedges and currency translation reserve)
	Adjusted RoE = Net income excluding earnings from SPPI-fail assets / Equity
	RoE = Operating profit after tax / Average adjusted shareholders' equity (adjusted for net unrealized gains/losses on assets and liabilities and cash flow hedges)

IFRS 17 impact on performance metrics

Leverage

Direction of change	Calculation approach
Majority within the population expect: Reduction in leverage ratio under IFRS 17 compared to IFRS 4 ratio (ranges from 1%pt to 11%pts), or no significant change	- Composites intend to include CSM in the denominator of the ratio (varying practices for tax and non-attributable expenses not in CSM) - Two composite groups intend to include risk adjustment in the denominator of the ratio as well

No explicit disclosure:



Examples:

Allianz	To include CSM (net of reinsurance) adj. for tax and non-attributable expenses not in CSM; IFRS 17 ratio at 23% (IFRS 4 at 26%) To continue measuring leverage based on SII, not IFRS (target < 30%) IFRS 17 ratio at 19% - 23% (IFRS 4 at 25% - 28%) IFRS 17 ratio at 16% (IFRS 4 at 26%) Munich RE IFRS 17 ratio at 10.7% (IFRS 4 at 14.7%)
tal anx.	To include both CSM and risk adjustment; IFRS 17 ratio at 22% (IFRS 4 at 33%)
ZURICH	To include both CSM and risk adjustment
HISCOX	IFRS 17 leverage ratio to decrease by 1%pt (vs IFRS 4)
ageas.	No material change in leverage expected (vs IFRS 4)

IFRS 17 impact on performance metrics

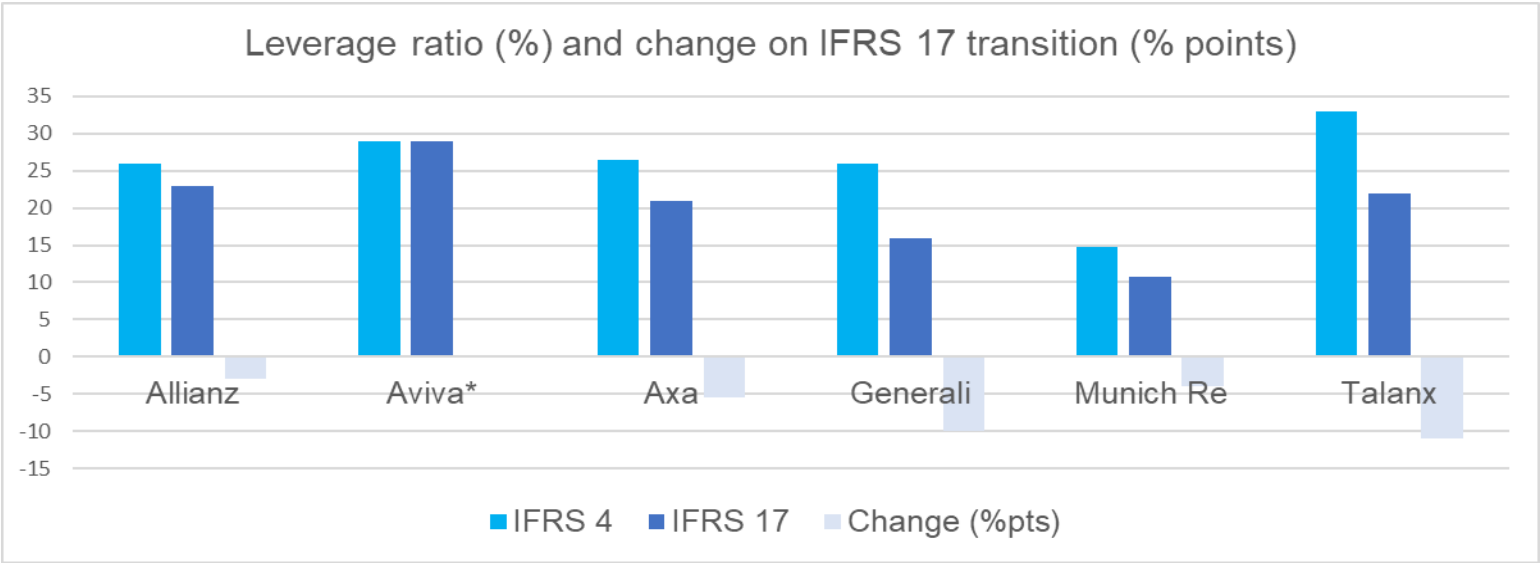
Leverage

Direction of change	Calculation approach
Majority within the population expect: Reduction in leverage ratio under IFRS 17 compared to IFRS 4 ratio (ranges from 1%pt to 11%pts), or no significant change	- Composites intend to include CSM in the denominator of the ratio (varying practices for tax and non-attributable expenses not in CSM) - Two composite groups intend to include risk adjustment in the denominator of the ratio as well

No explicit disclosure:



Examples:



* Measured on a SII basis

IFRS 17 impact on performance metrics

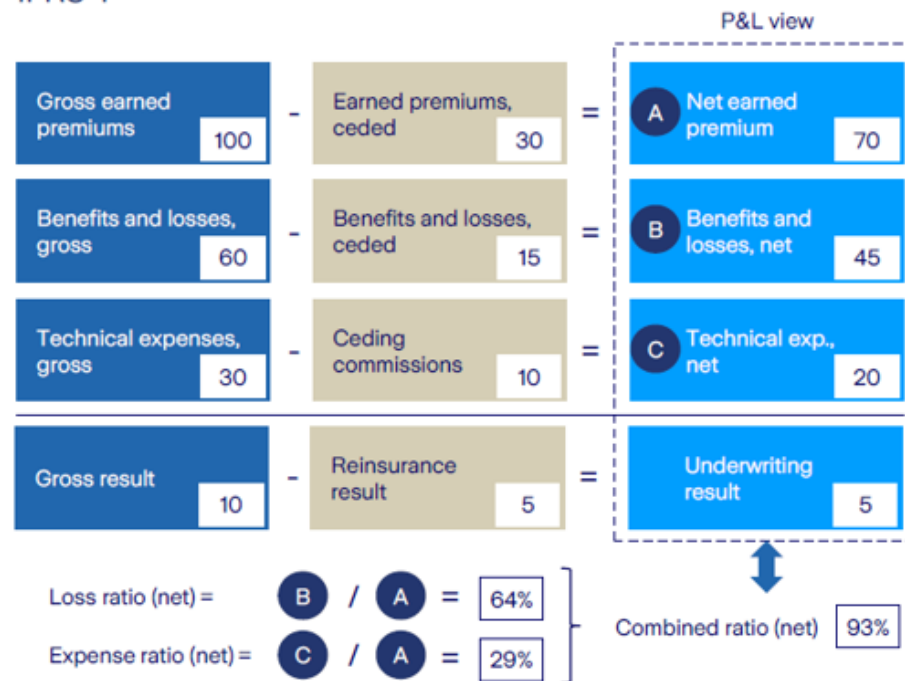
(1 of 2)

Combined ratio, Loss ratio, Prior period development ratio (PPD), Expense ratio

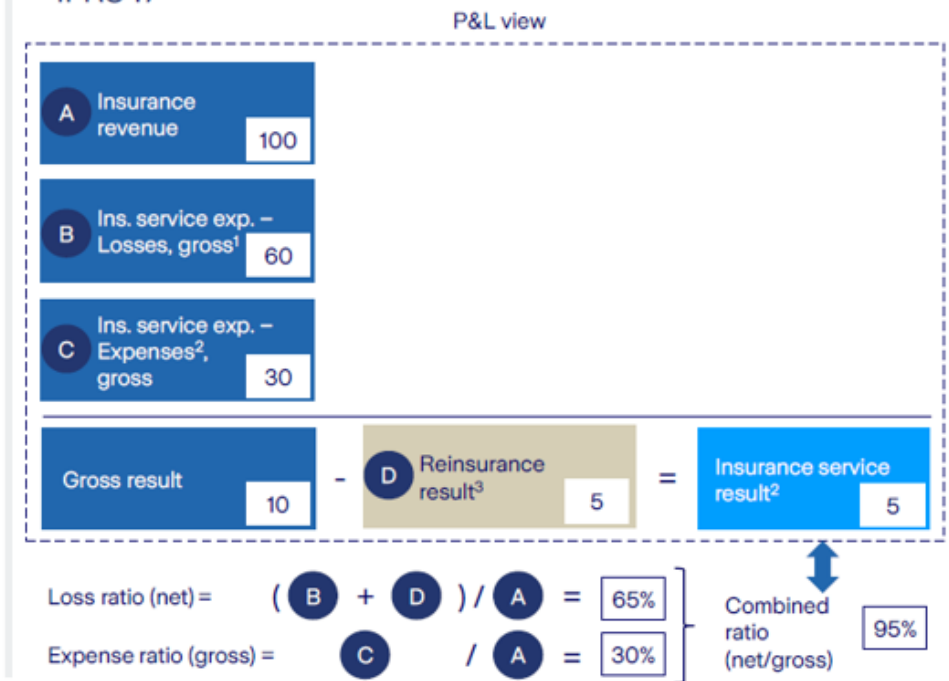
Prevalent choices in the population

- Choice of calculation basis (net or gross insurance revenue) appears to be driven by the business steering approach
- Practices vary for the treatment on non-attributable expenses (general admin expenses / overheads)
- No intention to continue reporting IFRS 4-basis ratios but plans to disclose GWP (as a non-IFRS KPI)

Illustration (extract): IFRS 4



IFRS 17



ILLUSTRATIVE NUMBERS

IFRS 17 impact on performance metrics

(2 of 2)

Combined ratio, Loss ratio, Prior period development ratio (PPD), Expense ratio

Net / Gross ratios					Net / Net ratios				Hybrid
	Allianz 	AXA 	GENERALI 	ZURICH 	Munich RE 	AVIVA 	DirectLine Group 	HISCOX 	Admiral Group PLC 
Loss ratio	▲◀▶	Not specified	▼	Not specified	▼	Not specified	Not specified	Not specified	◀▶
PPD ratio	Not specified	◀▶	▲	Not specified	◀▶	Not specified	Not specified	Not specified	Not specified
Expense ratio	▼ *	Not specified	▲ *	▼ *	▼ #	*	Not specified #	Not specified #	◀▶ #
Combined ratio	▼	▼	▲	▼▲	▼	◀▶	Not specified	▲	Not specified

Legend

▲	increase
▼	decrease
◀▶	no material change

- * Ratio includes attributable & non-attributable expenses
- # Ratio includes attributable expenses only

Hybrid ratios are on a basis net of XoL reinsurance and gross of QS reinsurance

Appendix

Illustrative disclosures

Discount rates (numerical)

The yield curves that were used to discount the estimates of future cash flows that do not vary based on the returns of the underlying items are as follows:

Product	Currency	20X4					20X3				
		1 year	5 years	10 years	20 years	30 years	1 year	5 years	10 years	20 years	30 years
Property and Casualty (issued and reinsurance held)	CU	3.02%	3.97%	4.56%	-	-	3.08%	4.03%	4.62%	-	-

Source: PwC, *IFRS 17, Insurance Contracts: An illustration*
Financial statements presentation and disclosures (extract)

The tables below set out the yield curves used to discount the cash flows of insurance contracts for major currencies.

	2023					2022				
	1 year	5 years	10 years	15 years	20 years	1 year	5 years	10 years	15 years	20 years
All other insurance contracts										
EUR	0.09%	0.26%	0.87%	1.42%	1.52%	0.06%	0.24%	0.85%	1.40%	1.50%
GBP	0.67%	1.61%	2.04%	2.24%	2.29%	0.62%	1.53%	1.91%	2.06%	2.18%
SGD	0.92%	1.79%	2.25%	2.48%	2.65%	0.81%	1.68%	2.20%	2.45%	2.63%
USD	1.04%	1.83%	2.18%	2.36%	2.48%	0.96%	1.78%	2.14%	2.34%	2.46%

Source: KPMG, *Illustrative disclosures for insurers*
Guide to annual financial statements IFRS 17 and IFRS 9 (extract)

Illustrative disclosures

Discount rates (narrative)

A mix of the bottom-up and top-down approaches was applied in the determination of the discount rates for different products.

The bottom-up approach was used to derive the discount rate for the cash flows that do not vary based on the returns on underlying items in the Participating contracts (excluding investment contracts without DPF that are not in the scope of IFRS 17). Under this approach, the discount rate is determined as the risk free yield adjusted for differences in liquidity characteristics between the financial assets used to derive the risk free yield and the relevant liability cash flows (known as an illiquidity premium). The risk free yield was derived using swap rates available in the market denominated in the same currency as the product being measured. When swap rates are not available, highly liquid sovereign bonds with a AAA credit rating were used. Management uses judgement to assess liquidity characteristics of the liability cash flows. Direct participating contracts and investment contracts with DPF are considered less liquid than the financial assets used to derive the risk free yield. For these contracts, the illiquidity premium was estimated based on market observable liquidity premium in financial assets adjusted to reflect the illiquidity characteristics of the liability cash flows.

The top-down approach was used to derive the discount rates for the cash flows that do not vary based on the returns on underlying items in all other contracts within the scope of IFRS 17. Under this approach, the discount rate is determined as the yield implicit in the fair value of a reference portfolio adjusted for differences between the reference portfolio of assets and respective liability cash flows. The reference portfolio comprises a mix of sovereign and corporate bonds available on the markets. The assets were selected in order to match the liability cash flows. The yield from the reference portfolio was adjusted to remove both expected and unexpected credit risk. These adjustments were estimated using information from observed historic levels of default and credit default swaps relating to the bonds included in the reference portfolio.

For both the bottom-up and top-down approaches, observable market information is available for up to 20 years. For the unobservable period, the yield curve was interpolated between an ultimate rate and the last observable point using the Smith-Wilson method.

Source: PwC, IFRS 17, Insurance Contracts: An illustration
Financial statements presentation and disclosures

Illustrative disclosures

Explanation of relationship of IFIE and investment return on assets

20X4	Life Risk	Savings	Participating			Property and Casualty	Other	Total
			Direct participating contracts	Investment contracts with DPF	Investment contracts without DPF			
Net investment income (expenses) - underlying assets								
Interest revenue from financial assets not measured at FVTPL	-	-	2,157	-	-	-	-	2,157
Net gains on FVTPL investments	-	1,311	-	204	756	-	-	2,271
Net credit impairment losses	-	-	(30)	-	-	-	-	(30)
Net gains on investments in debt securities measured at FVOCI	-	-	464	-	-	-	-	464
Net investment income - underlying assets	-	1,311	2,591	204	756	-	-	4,862
Net investment income (expenses) - other investments								
Interest revenue from financial assets not measured at FVTPL	-	-	-	-	-	-	539	539
Net gains on FVTPL investments	1,172	1,602	-	-	-	1,532	4,552	8,858
Net gains from the derecognition of financial assets measured at AC	-	-	-	-	-	-	22	22
Net credit impairment losses	-	-	-	-	-	-	(10)	(10)
Net gains on investments in debt securities measured at FVOCI	-	-	-	-	-	-	40	40
Net investment income - other investments	1,172	1,602	-	-	-	1,532	5,143	9,449
Net investment income (expenses) - other								
Net change in investment contract liabilities	-	-	-	-	(756)	-	-	(756)
Net gains from fair value adjustments to investment properties	-	-	-	-	-	-	157	157
Net investment income (expenses) - other	-	-	-	-	(756)	-	157	(599)
Total net investment income	1,172	2,913	2,591	204	-	1,532	5,300	13,712

20X4	Life Risk	Participating				Property and Casualty	Other	Total
		Savings	Direct participating contracts	Investment contracts with DPF	Investment contracts without DPF			
Finance income (expenses) from insurance contracts issued								
Changes in fair value of underlying assets of contracts measured under the VFA	-	-	(2,164)	-	-	-	-	(2,164)
Interest accreted	(2,230)	(1,356)	(29)	(423)	-	(443)	-	(4,481)
Effect of changes in interest rates and other financial assumptions	1,388	958	3	(22)	-	13	-	2,340
Effect of changes in FCF at current rates when CSM is unlocked at locked-in rates	17	27	-	1	-	-	-	45
Foreign exchange differences	-	-	-	143	-	-	-	143
Finance expenses from insurance contracts issued	(825)	(371)	(2,190)	(301)	-	(430)	-	(4,117)
Finance income (expenses) from reinsurance contracts held								
Interest accreted	1,484	-	-	-	-	27	-	1,511
Effect of changes in interest rates and other financial assumptions	(998)	-	-	-	-	-	-	(998)
Effect of changes in FCF at current rates when CSM is unlocked at locked-in rates	(12)	-	-	-	-	-	-	(12)
Finance income from reinsurance contracts held	474	-	-	-	-	27	-	501
Net insurance finance expenses	(351)	(371)	(2,190)	(301)	-	(403)	-	(3,616)

Source: PwC, IFRS 17, Insurance Contracts: An illustration
Financial statements presentation and disclosures

Illustrative disclosures

Sensitivities – interest rates

	0.3% increase in interest rates Impact on:						0.3% decrease in interest rates Impact on:			
	Net insurance/ investment contracts balance	Investment assets subject to interest rate risk ⁽¹⁾	Net insurance/ investment contracts balance	Investment assets	Profit or loss	Equity	Net insurance/ investment contracts balance	Investment assets	Profit or loss	Equity
31 December 20X4										
Life Risk	(4,278)	4,407	(88)	(103)	(191)	(140)	91	107	198	146
Property and Casualty	(17,604)	18,132	68	(109)	(41)	(30)	(70)	114	44	32

The analysis is based on a change in an assumption while holding all other assumptions constant. In practice, this is unlikely to occur, and changes in some of the assumptions may be correlated. No changes were made by the Group in the methods and assumptions used in preparing the above analysis.

Source: PwC, IFRS 17, Insurance Contracts: An illustration
Financial statements presentation and disclosures (extract)

Illustrative disclosures

Risk adjustment

(1 of 2)

The risk adjustments for non-financial risk are determined using the following techniques:

- *non-life contracts*: a confidence level technique;
- *life and participating contracts outside Europe*: a confidence level technique; and
- *life and participating contracts in Europe*: a cost of capital technique.

To determine the risk adjustments for non-financial risk for reinsurance contracts, the Group applies these techniques both gross and net of reinsurance and derives the amount of risk being transferred to the reinsurer as the difference between the two results.

Applying a confidence level technique, the Group estimates the probability distribution of the expected present value of the future cash flows from insurance contracts at each reporting date and calculates the risk adjustment for non-financial risk as the excess of the value at risk at the 75th percentile (the target confidence level) over the expected present value of the future cash flows.

Source: KPMG, *Illustrative disclosures for insurers*
Guide to annual financial statements
IFRS 17 and IFRS 9

Illustrative disclosures

Risk adjustment

(2 of 2)

Applying a cost of capital technique, the Group determines the risk adjustment for non-financial risk by applying a cost-of-capital rate to the amount of capital required for each future reporting date and discounting the result using risk-free rates adjusted for illiquidity. The required capital is determined by estimating the probability distribution of the present value of future cash flows from insurance contracts at each future reporting date and calculating the capital that the Group would require to meet its contractual obligations to pay claims and expenses arising over the duration of the contracts at a 90% confidence level. The cost-of-capital rate represents the additional reward that investors would require for exposure to the non-financial risk. The Group's weighted-average cost-of-capital rate is 4.5% (2022: 4.5%).

The risk adjustments for non-financial risk for life and participating contracts in Europe correspond to the following confidence levels.

	2023		2022	
	Gross of reinsurance	Net of reinsurance	Gross of reinsurance	Net of reinsurance
France	75.4%	80.6%	75.9%	81.0%
Germany	78.3%	83.8%	78.6%	84.0%
UK	76.7%	81.9%	76.9%	82.1%

Source: KPMG, Illustrative disclosures for insurers
Guide to annual financial statements IFRS 17 and IFRS 9

Illustrative disclosures

Liquidity

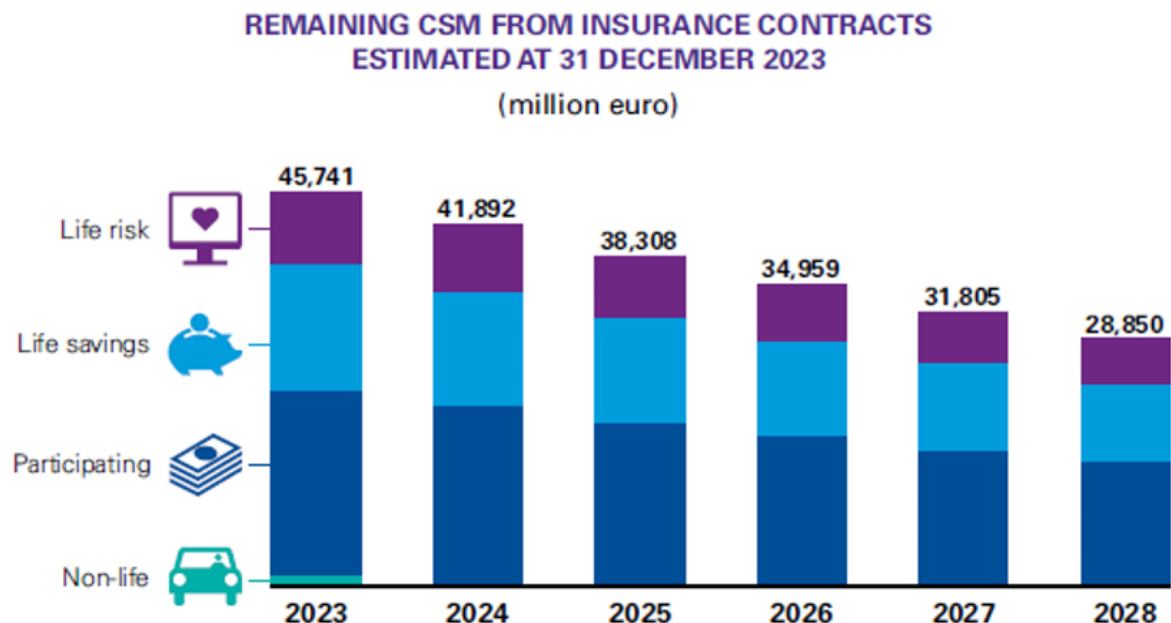
The following tables present the estimated amount and timing of the remaining contractual discounted cash flows arising from investment assets and insurance liabilities in the Life Risk and Savings product lines. When debt securities mature, the proceeds not needed to meet liability cash flows will be reinvested.

31 December 20X4	1	2	3	4	5	6-10	>10	Total
Underlying assets ⁽¹⁾								
Government bonds	14	12	20	22	33	130	1,209	1,440
Other debt securities	106	84	148	158	243	950	8,867	10,556
	120	96	168	180	276	1,080	10,076	11,996
Other investments								
Government bonds	194	155	271	290	445	1,743	16,268	19,366
Other debt securities	119	95	166	178	273	1,068	9,971	11,870
	313	250	437	468	718	2,811	26,239	31,236
Total investment assets	433	346	605	648	994	3,891	36,315	43,232
Insurance contract balances								
Insurance contract assets	5,235	4,374	3,592	3,010	2,482	7,317	583	26,593
Reinsurance contract assets	(14,153)	(11,791)	(9,726)	(7,853)	(6,507)	(20,633)	4,803	(65,860)
Insurance contract liabilities	45,852	40,458	35,645	31,260	27,807	36,445	(129,311)	88,156
Reinsurance contract liabilities	(3,872)	(3,230)	(2,647)	(2,214)	(1,820)	(5,316)	(237)	(19,336)
Total insurance contract balances	33,062	29,811	26,864	24,203	21,962	17,813	(124,162)	29,553
Net discounted cash flows	33,495	30,157	27,469	24,851	22,956	21,704	(87,847)	72,785

Source: PwC, IFRS 17, Insurance Contracts: An illustration
Financial statements presentation and disclosures

Illustrative disclosures

CSM: expected allocation pattern



The following table sets out when the Group expects to recognise the remaining CSM in profit or loss after the reporting date for contracts not measured under the PAA.^a

In millions of euro

	1 year or less	1-2 years	2-3 years	3-4 years	4-5 years	5-10 years	More than 10 years	Total
31 December 2023								
Insurance contracts								
Life risk	538	479	426	376	329	1,076	3,463	6,687
Life savings	987	939	891	862	823	3,785	3,595	11,882
Participating	2,311	2,165	2,032	1,916	1,803	9,314	7,617	27,158
Non-life	13	1	-	-	-	-	-	14
	3,849	3,584	3,349	3,154	2,955	14,175	14,675	45,741
Reinsurance contracts								
Life risk	(8)	(7)	(7)	(7)	(6)	(17)	(58)	(110)
Participating	(24)	(22)	(21)	(19)	(18)	(81)	(68)	(253)
	(32)	(29)	(28)	(26)	(24)	(98)	(126)	(363)

Source: KPMG, Illustrative disclosures for insurers
Guide to annual financial statements IFRS 17 and IFRS 9



Guy Carpenter & Company LLC provides this presentation for general information only. The information contained herein is based on sources we believe reliable, but we do not guarantee its accuracy, and it should be understood to be general insurance/reinsurance information only. Guy Carpenter & Company LLC makes no representations or warranties, express or implied. The information is not intended to be taken as advice with respect to any individual situation and cannot be relied upon as such. Please consult your insurance/reinsurance advisors with respect to individual coverage issues.

Statements concerning tax, accounting, legal or regulatory matters should be understood to be general observations based solely on our experience as reinsurance brokers and risk consultants, and may not be relied upon as tax, accounting, legal or regulatory advice, which we are not authorized to provide. All such matters should be reviewed with your own qualified advisors in these areas.

Readers are cautioned not to place undue reliance on any historical, current or forward-looking statements. Guy Carpenter & Company LLC undertakes no obligation to update or revise publicly any historical, current or forward-looking statements, whether as a result of new information, research, future events or otherwise.

This document or any portion of the information it contains may not be copied or reproduced in any form without the permission of Guy Carpenter & Company LLC, except that clients of Guy Carpenter & Company LLC need not obtain such permission when using this report for their internal purposes.

The trademarks and service marks contained herein are the property of their respective owners.
© 2023 Guy Carpenter & Company LLC

This presentation must not be shared with third parties without our prior written consent.